

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think> To solve , first calculate the distance John initially travels to the store . Then calculate the distance he travels back home with traffic , and finally , calculate the distance he travels after the traffic in the last half-hour and the last half-hour of the 4 -hour drive . Sum the distance he is away from home at the end of the 4 hours . Distance = speed $\times$ time . For the first part , he travels 3 hours at 60 mph , so $3 * 60 = 180$ miles . For the return trip , he spends 2 hours in traffic , so he doesn't cover any distance during that time . In the next half-hour , he travels at 30 mph , so $0.5 * 30 = 15$ miles . For the last half-hour of the 4 -hour return , he travels at 80 mph , so $0.5 * 80 = 40$ miles . Adding these distances , 180 (to the store) - 15 (first part of the return trip) + 40 (last part of the return trip) = 205 miles away from home at the end of the 4 hours . </think><answer>205</answer>

